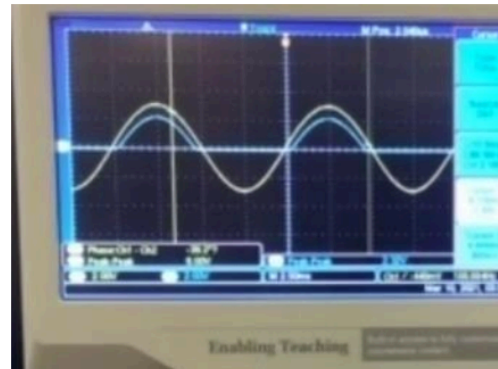
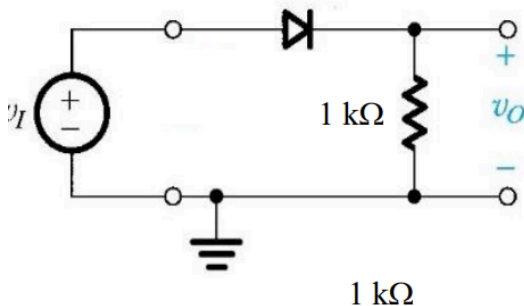


- Experiment #1: Half Wave Rectifier
- Experiment #2: AC to DC Converter
- Experiment #3: Limiting and Protection Circuit
- Experiment #4: I(V) characteristics of a p-n Zener diode
- Experiment #5: Zener Voltage Regulator
- Experiment #6: Regulated DC power supply

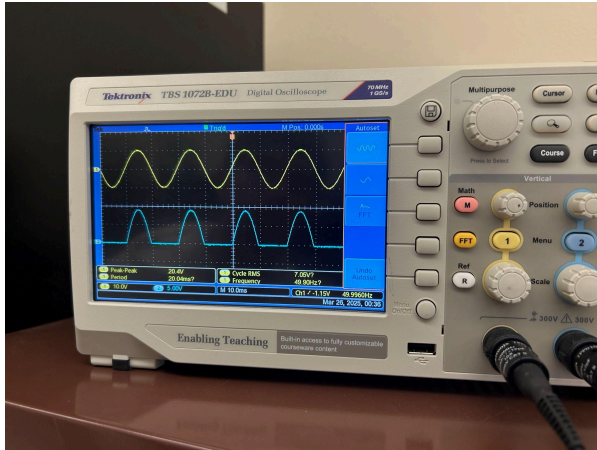
Experiment #1: Half-Wave Rectifier: (Simulation + Lab Measurement)

Construct the following Half-Wave Rectifier circuit.

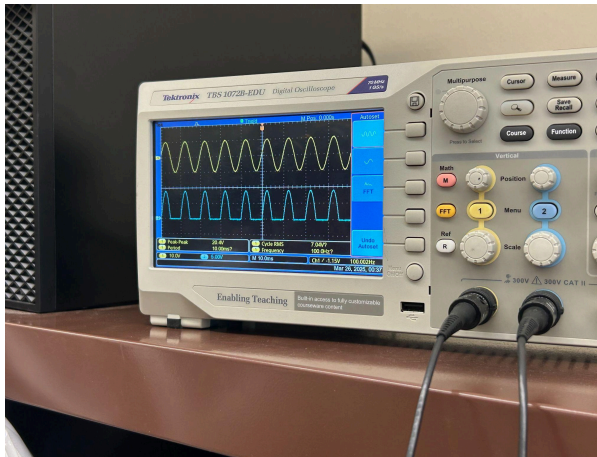


1. Let the source be your sinusoidal signal generator. Set the frequency to be 50 Hz and the amplitude to be 20 V_{pk-pk} (peak to peak).
2. Using the oscilloscope observe and record the waveforms v_I and v_O . Take clear photos of the oscilloscope screen displaying both waveforms as per the example above. Ensure that the photos clearly display the x- and y- scales of the waveforms.
3. Repeat step 2 for a frequency of 100 Hz and 1000 Hz.

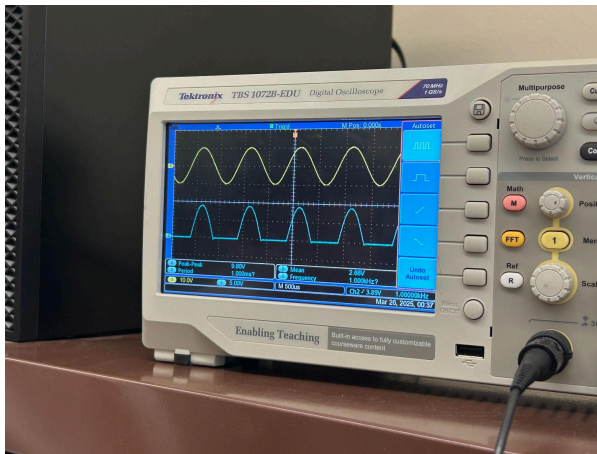
R=0.98829KOHM



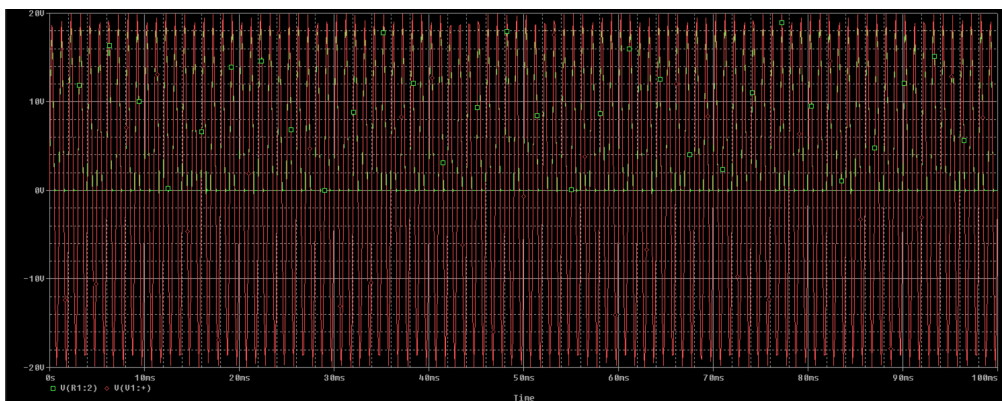
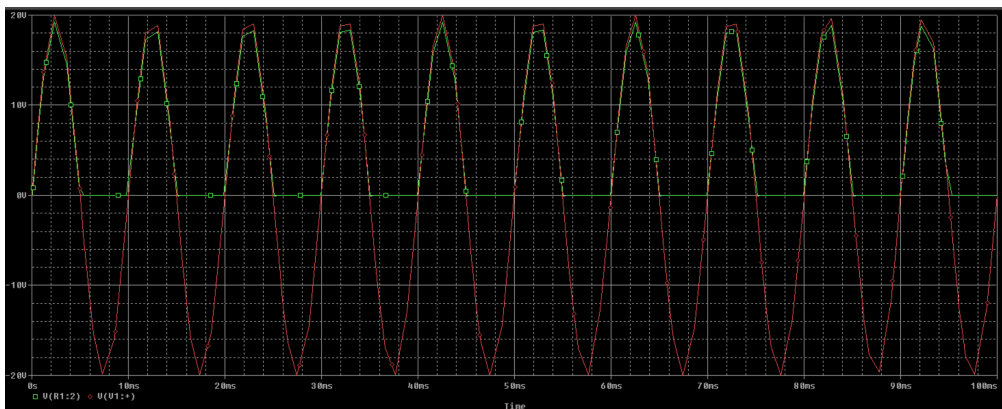
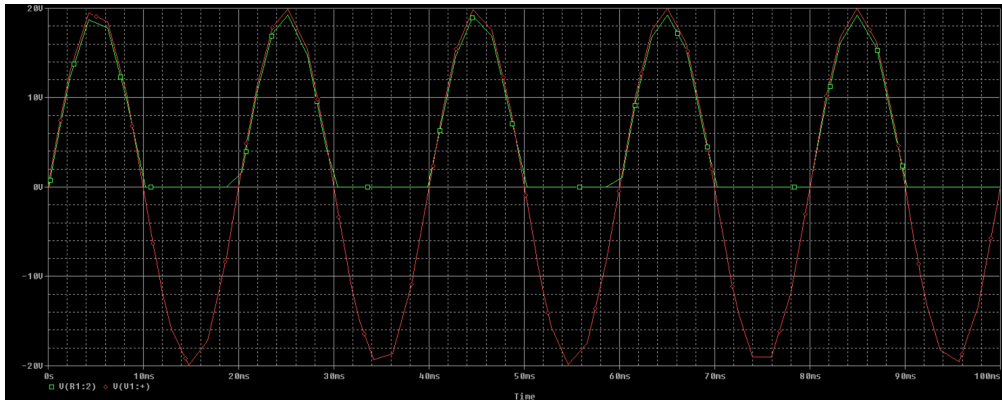
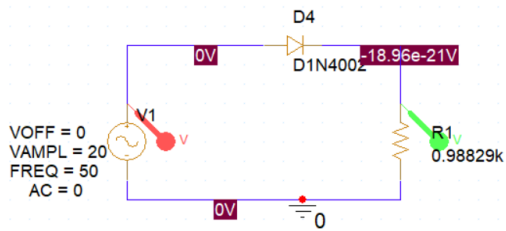
50Hz



100Hz



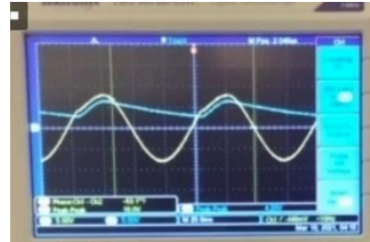
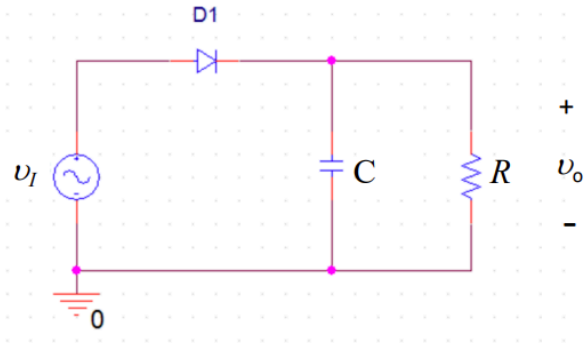
1000Hz



Conclusion: I believe that this experiment was successful due to input and output voltages from the lab measurement and simulation look similar at different frequencies.

Experiment #2: AC to DC converter (Simulation + Lab Measurement)

Build the circuit below. Let the source be your sinusoidal signal generator set to frequency of 50Hz and 20-Vpk-pk (peak to peak) amplitude.

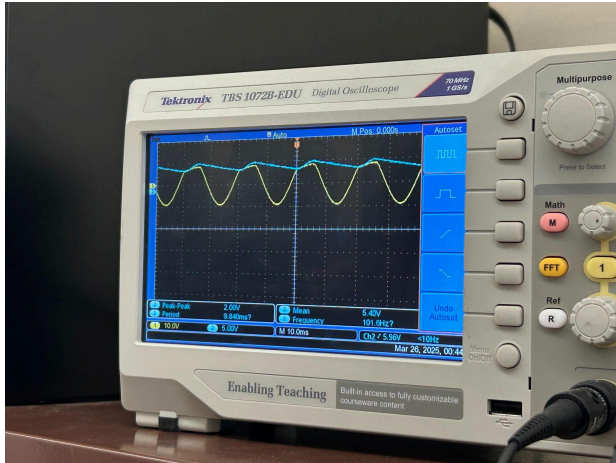


1. Let $C = 100 \mu\text{F}$ and $R = 470 \text{ ohms}$. Measure and record the values of these two components
 $C = \underline{\hspace{2cm}}$ $R = \underline{\hspace{2cm}}$
2. Using the oscilloscope observe and record the waveforms v_I and v_o . Take clear photos of the oscilloscope screen displaying both waveforms as per the example above. Ensure that the photos clearly display the x- and y- scales of the waveforms.
3. Let C be same and R be higher e.g. 2000 ohms. Measure and record new R value: $R = \underline{\hspace{2cm}}$
4. Repeat step 2 for the new value of R .

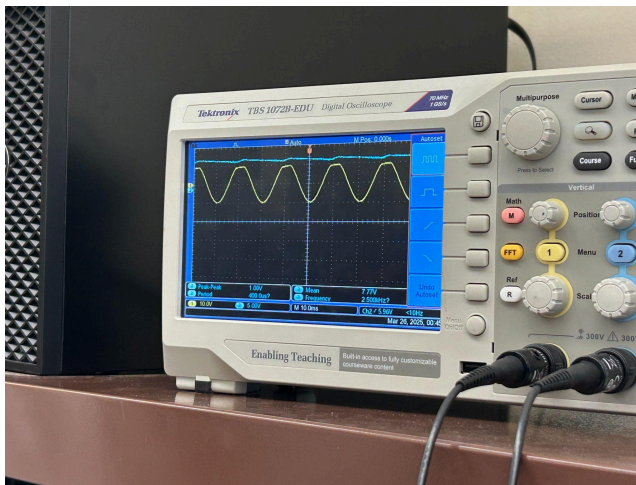
$C = 97.84 \text{ microFarad}$

$R = 479.5 \text{ ohm}$

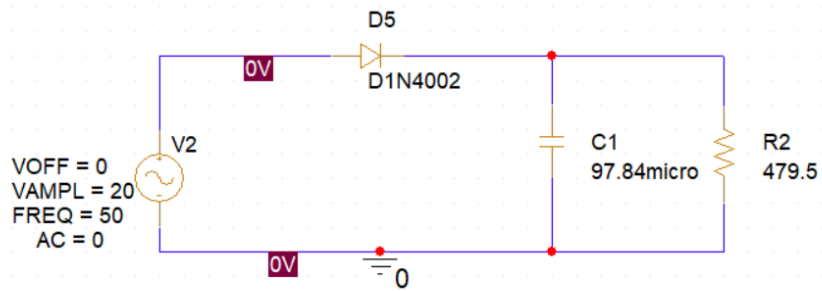
$R = 2.182 \text{ KOHM}$

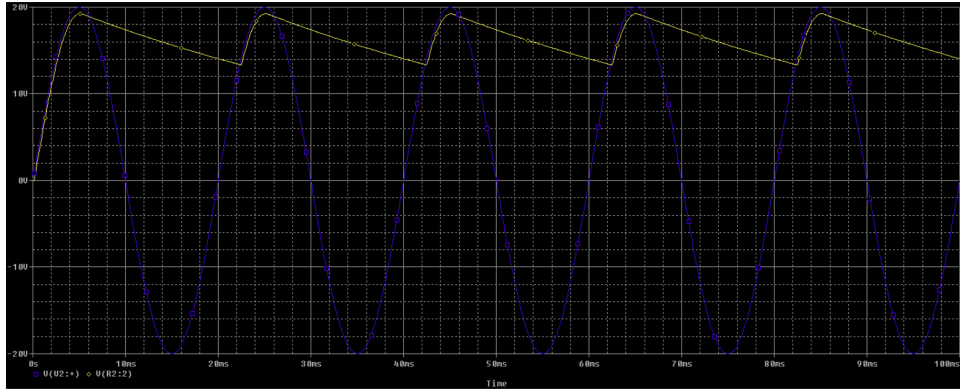


479.5ohm

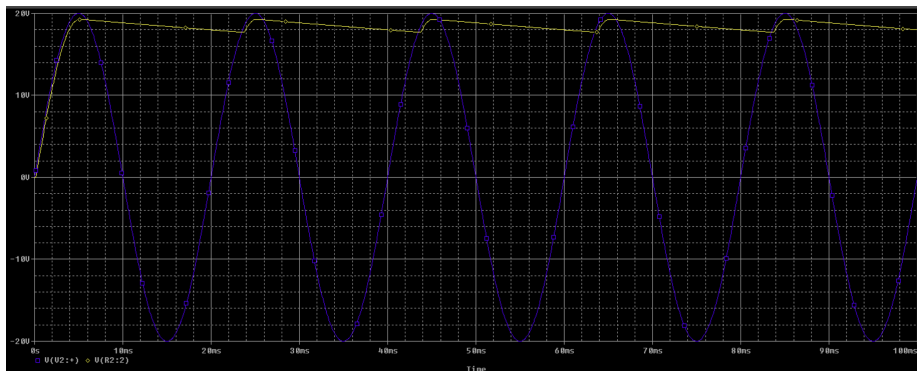
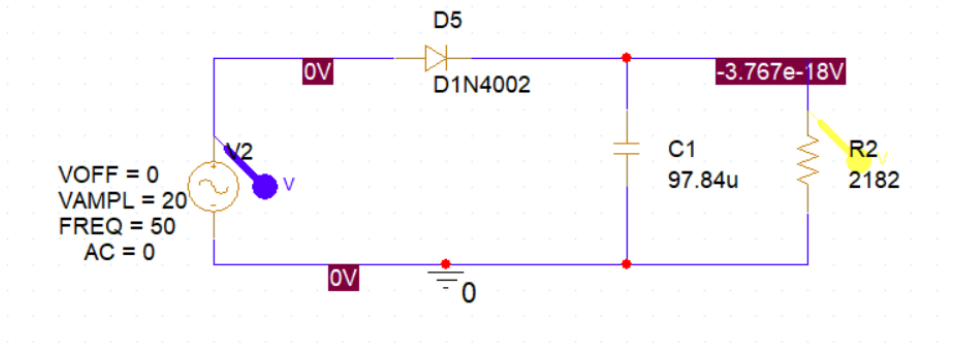


2.182Kohm





479.5ohm

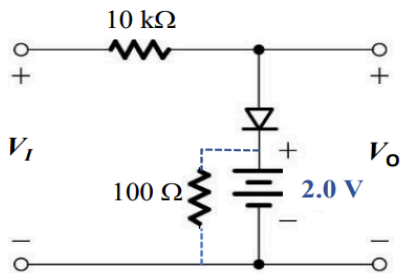


2.182Kohm

Conclusion: I also believe that this experiment was successful because in both the lab measurements and simulation you were able to see how the increase in resistances would make the output voltage shift up over the input voltage.

Experiment #3: Limiting and Protection Circuits (Simulation + Lab Measurement)

Construct the following circuit and measure the output voltage as a function of input voltage. Note that if the 2 V voltage source cannot sink current then place a 100 ohm resistor in parallel to 2 V voltage source as shown with dotted line. Complete the Table below.

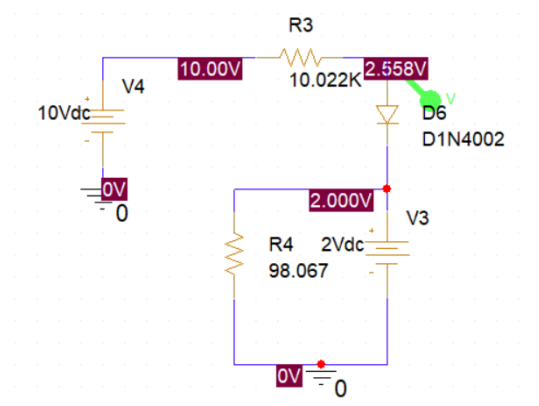


V_I (V)	V_O (V)
-5	
-4	
-3	
-2	
-1	
0	
+1	
+2	
+3	
+4	
+5	
+10	

R=10.022KOHM

R=98.067OHM

Experiment 3		
	V_i (V)	V_o (V)
	-5	-4.8786
	-4	-3.9316
	-3	-3.0833
	-2	-2.0409
	-1	-1.0925
	0	0.536mV
	1	0.99566
	2	2.0063
	3	2.4901
	4	2.5293
	5	2.5555
	10	2.6067



Experiment 3 Simulated		
	$V_i(V)$	$V_o(V)$
	-5	-5
	-4	-4
	-3	-3
	-2	-2
	-1	-1
	0	0.1431micro(V)
	1	1
	2	2
	3	2.426
	4	2.476
	5	2.502
	10	2.558

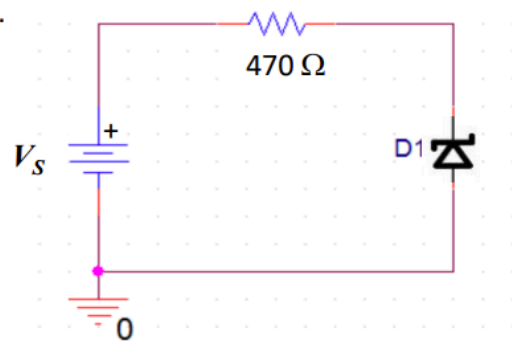
Conclusion: This experiment was a success due to the close correlation between the output voltage in the lab measurement and the simulation. The output voltages were nearly identical at each input voltage in the table.

Experiment #4: I(V) characteristics of a Zener p-n diode (Simulation + Lab Measurement)

Set-up the measuring circuit shown below to determine the characteristics of a Zener p-n diode in the REVERSE region:

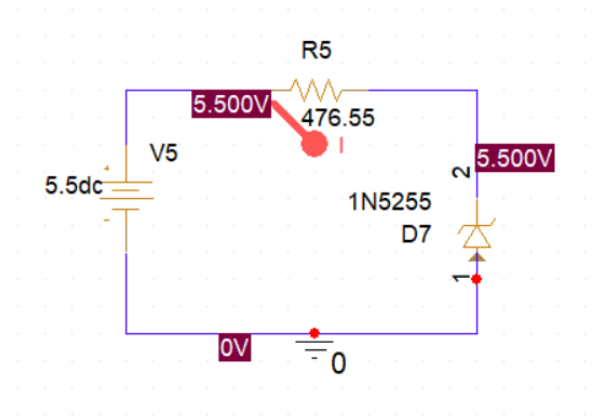
- Vary the voltage of the voltage source starting from 0 V and increase it by +1V step. At each value of the voltage source measure the current that flows through the resistor and the voltage across the Zener diode. Continue increasing the voltage source until you reach a value of 5 volt higher than the breakdown region. The Zener diode breakdown occurs when a sudden increase of current is detected.
- Within +/-1 V from the breakdown voltage take additional data, by varying the voltage at 0.1V increment and measure the current flowing through the resistor and the voltage across the Zener diode.
- Record the I(V) characteristics of your Zener diode using the curve tracer. Record the Zener breakdown voltage value:

$V_Z =$ _____



476.55 OHM

Experiment 4			
V_i (V)	I (A)	V_i (V)	I (mA)
0	0	5.5	0
1	0	5.6	0.0001
2	0	5.7	0.0001
3	0	5.8	0.0002
4	0	5.9	0.0002
5	0	6	0.0003
6	0	6.1	0.0004
7	1.194mA	6.2	0.0007
8	3.2381mA	6.3	0.001
9	5.3747mA	6.4	0.0017
10	7.4290mA	6.5	0.1154
11	9.3371mA	6.6	0.3627
12	11.318mA	6.7	0.5079
		6.8	0.7068
		6.9	0.8843
		7	1.0792
		7.1	1.3722
		7.2	1.5051
		7.3	1.7449
		7.4	1.9885
		7.5	2.1519

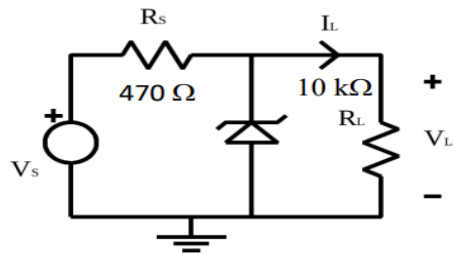


Experiment 4 Simulated				
Vi (V)	I (A)			I(nA)
0	0		5.5	57.574nA
1	32.057nA		5.6	57.989nA
2	39.463nA		5.7	58.404nA
3	45.515nA		5.8	58.819nA
4	50.744nA		5.9	59.234nA
5	55.407nA		6	59.649nA
6	59.649nA		6.1	60.038nA
7	63.535nA		6.2	60.426nA
8	67.214nA		6.3	60.815nA
9	70.645nA		6.4	61.203nA
10	73.891nA		6.5	61.592nA
11	76.978nA		6.6	61.981nA
12	79.926nA		6.7	62.369nA
			6.8	62.758nA
			6.9	63.146nA
			7	63.535nA
			7.1	63.924nA
			7.2	64.312nA
			7.3	64.701nA
			7.4	65.089nA
			7.5	65.478nA

Conclusion: This lab was not a success due to inconsistencies in the lab measurements and the simulation results. I believe that this issue came from not using the proper diode in the lab experiment causing the data to be so far off.

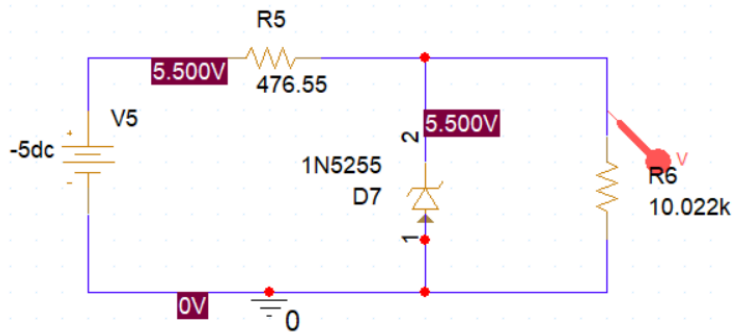
Experiment #5: Zener Voltage Regulator (Simulation + Lab Measurement)

In the circuit from Experiment #4, add a 10 kΩ Load Resistor R_L . Set the voltage source at a voltage value equal to the Zener breakdown voltage minus five volts, e.g. if Zener breakdown is 10 V set your source at 5V. Vary the voltage of the voltage source starting from the above initial value and increase it by +1V, until you reach a value of 5 V higher than the Zener breakdown. At each source voltage value measure the current flowing through the load resistor R_L and the voltage across the load resistor R_L . Make a Table of your measurements.



V_s (V)	V_L (V)	I_L (mA)
$V_z - 5$		
$V_z - 4$		
$V_z - 3$		
$V_z - 2$		
$V_z - 1$		
V_z		
$V_z + 1$		
$V_z + 2$		
$V_z + 3$		
$V_z + 4$		
$V_z + 5$		

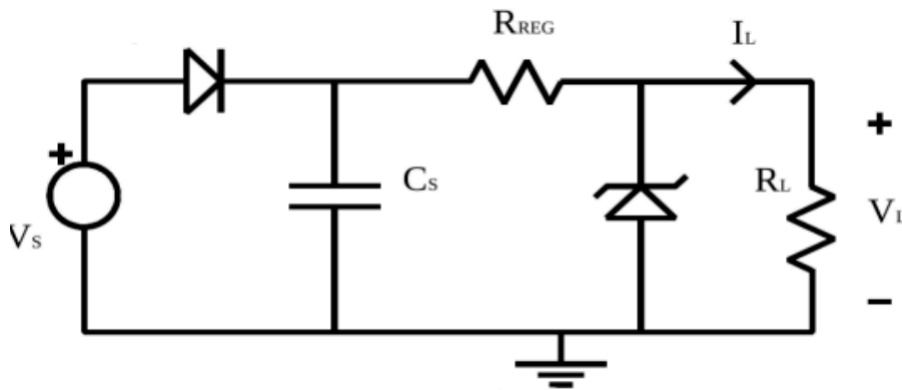
Experiment 5			
	V_s (V)	V_L (V)	I_L (mA)
	V_z-5	1.3731V	0.1346mA
	V_z-4	2.3439V	0.2331mA
	V_z-3	3.309V	0.3299mA
	V_z-2	4.2206V	0.4228mA
	V_z-1	5.2105V	0.5246mA
	$V_z=6.4V$	6.1674V	0.6201mA
	V_z+1	6.5677V	0.655mA
	V_z+2	6.5766V	0.6561mA
	V_z+3	6.5845V	0.6557mA
	V_z+4	6.5919V	0.6564mA
	V_z+5	6.5951V	0.657mA



Experiment 5 Simulated			
	Vs(V)	VL(V)	IL(A)
	Vz-5	-801.6mV	-79.988microA
	Vz-4	-788.65mV	-78.693microA
	Vz-3	-770.89mV	-76.667microA
	Vz-2	-737.75mV	-73.613microA
	Vz-1	-640.88mV	-63.948microA
	Vz=6.4V	6.1095V	609.605microA
	Vz+1	0.954V	95.250microA
	Vz+2	1.909V	190.501microA
	Vz+3	2.863V	285.752microA
	Vz+4	3.818V	381.003microA
	Vz+5	4.773V	476.254microA

Conclusion: This lab was also another failure for the same reason from experiment 4, the issue came from not using the proper diode in the lab experiment causing the data to be so far off.

Experiment #6: Regulated DC power supply: (Simulation + Lab Measurement)

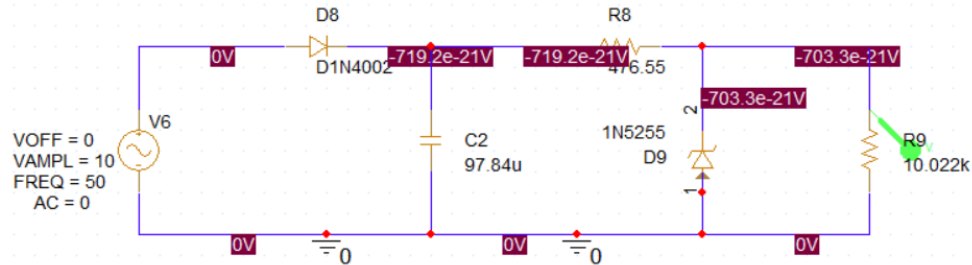


In the circuit from Experiment 1, add Experiment 5 instead of the load resistance to regulate the AC voltage supply to the DC output. Build the final circuit using 470 ohm, $C_s = 100 \mu\text{F}$, R_L should be a minimum of 1.8k ohm. Vary the V_s and observe V_L .

$R_{\text{reg}} = 470 \text{ ohm}$

$R_L = 2.2 \text{ kohm}$

Experiment 6			
	Vs		Vout
	1 Vpp		0.19815V
	2 Vpp		0.65987V
	3 Vpp		1.13533V
	4 Vpp		1.5964V
	5 Vpp		2.0555V
	6 Vpp		2.6351V
	7 Vpp		3.1092V
	8 Vpp		3.5073V
	9 Vpp		4.5952V
	10 Vpp		4.6432V



Conclusion: This lab was also a failure not because of lab measurement values but our simulation values. In PSpice I was supposed to get a DC voltage on the output but instead I got an exponential value that does not satisfy our lab results.